

Data Analytics Practical 1 – Viva Questions & Answers with Code

Complete Code:

```
import pandas as pd
df = pd.read_csv("dataset_facebook.csv", delimiter=";")

subset1 = df[['Type', 'Category', 'Post Month']]
subset2 = df[['Type', 'Category', 'comment']]

merged = pd.concat([subset1, subset2], axis=1)
sorted_data = merged.sort_values(by=['comment'], ascending=False)

transposed = merged.transpose()
shape = df.shape

import numpy as np
arr = np.array(df[['Post Month', 'Post Weekday']].head(6))
reshaped = arr.reshape(3, 4)

print(subset1.head())
print(subset2.head())
print(sorted_data.head())
print(transposed)
print("Shape:", shape)
print("Reshaped Array:", reshaped)
```

Viva Questions and Answers:

Q: What is Pandas?

A: A Python library used for data analysis and manipulation.

Q: What is a DataFrame?

A: A 2D table-like structure with rows and columns.

Q: Why use read_csv()?

A: To load CSV file into a DataFrame.

Q: What is delimiter='; '?

A: Specifies semicolon as column separator.

Q: What is subsetting?

A: Selecting specific columns or rows.

Q: Difference between df['col'] and df[['col']]?

A: First gives Series, second gives DataFrame.

Q: What is pd.concat()?

A: Used to combine DataFrames.

Q: Meaning of axis=1?

A: Column-wise joining.

Q: What is sorting?

A: Arranging data in ascending or descending order.

Q: What is transpose?

A: Swapping rows and columns.

Q: What does df.shape return?

A: Number of rows and columns.

Q: What is NumPy?

A: Library for numerical computations.

Q: What is reshape?

A: Changing dimensions of array.

Q: Condition for reshape?

A: Total elements must remain same.

Q: What is head()?

A: Displays first 5 rows.

Q: What is KeyError?

A: Error when column name is incorrect.